FDI Income by Economic Activity, BMD4 Preview Report and Planning

This dataset has been obtained from the OECD (Link: https://data-explorer.oecd.org/vis?lc=en&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_FDI%40DF_FDI_INC_IND&df[ag]=OECD.DAF.INV&dq=.T_D4P_F.USD_EXC.....W.IMC._T%2BA_B%2BC%2BC10T12%2BC13T18X15%2BC19T22%2BC24T28X27%2BC29_30%2BC15_23_27_31T33%2BD%2BE%2BF%2BGTU%2BG%2BH%2BI%2BJ%2BK%2BL%2BM%2BN%2BO_T_U%2BP%2BQ%2BR%2BS%2B_X.A.&lom=LASTNPERIODS&lo=2&to[TIME_PERIOD]=false).

The dataset provides various information about the foreign direct investments received by countries from one another, including the sectors in which they are made, their size, geographical distribution, etc. In this context, 24 variables were observed, resulting in a total of 10,344 observations.

Of these 24 variables, 6 represent OECD's internal coding/structure identifiers and do not contribute to the analysis (as they take the same value for every single observation). Therefore:

The types of the obtained variables are as follows:

1. STRUCTURE: Type of data structure → Meaningless / Removed
2. STRUCTURE_ID: Technical identifier of the data structure → Meaningless / Removed
3. STRUCTURE_NAME: Name of the data structure → Meaningless / Removed
4. ACTION: Type of operation on the data (usually "I" – Insert) → Meaningless / Removed
5. REF_AREA: Reference area or country code (e.g., USA, TUR) → Categorical
6. MEASURE: Type of measured variable → Categorical
7. UNIT_MEASURE: Unit of measurement (e.g., USD) → Categorical
8. MEASURE_PRINCIPLE: Measurement principle (Asset/Liability or Directional) → Categorical
9. ACCOUNTING_ENTRY: Accounting entry type (Net, Gross, etc.) → Categorical
10. TYPE_ENTITY: Type of entity → Categorical
11. FDI_COMP: FDI component → Categorical
12. SECTOR: Institutional sector in which the investment is made → Categorical
13. COUNTERPART_AREA: Counterpart country or region → Categorical
14. LEVEL_COUNTERPART: Level of the counterpart (World, Region, etc.) → Categorical
15. ACTIVITY: Economic activity branch (sectoral breakdown) → Categorical
16. FREQ: Data frequency (A: Annual, Q: Quarterly) → Categorical

17. FDI_COLLECTION_ID: Data collection method/source → Categorical

18. TIME_PERIOD: Time period (Year) → Numeric (Date)

19. OBS_VALUE: Observation value (main numerical data) → Numeric

20. OBS_STATUS: Observation status (Official, Estimated, etc.) → Categorical

21. UNIT_MULT: Unit multiplier (e.g., 6: Millions) → Numeric

22. CONF_STATUS: Confidentiality status → Meaningless / Removed

23. CURRENCY: Currency used → Categorical

24. DECIMALS: Number of decimal places → Meaningless / Removed

Even after removing the 6 variables, the variables **CURRENCY** and **UNIT_MULT** — which have not yet been examined in detail but appear uniform at first glance — also show no variation. If no differences are observed during SQL-based data manipulation, they will be marked as meaningless and removed depending on the context.

If no logical inconsistencies are observed in the SQL environment, the analysis will proceed directly in the Python/Jupyter Notebook environment.

In the Python/Jupyter Notebook environment, descriptive statistics will first be performed using the meaningful variables to become familiar with the data. Next, the distributions of numerical variables will be examined through graphs and normality tests to determine the type of distribution. Then, relationships/dependencies between variables will be identified:

- Categorical × Categorical → Chi-square
- Categorical × Numerical → ANOVA
- Numerical × Numerical → Correlation

Afterward, hypothesis testing will be conducted.

Hypotheses will be decided after observing the data, and inferential conclusions will be drawn accordingly. Then, modeling will begin.

A critical indicator for one of the key inferential statistics in modeling is the **OBS_STATUS** variable. This variable explains the reliability of **OBS_VALUE**. Depending on the detected relationships and distributions, modeling will be carried out to reach values with status **A** (official/actual/normal observation) within sufficient confidence intervals. If no statistically significant results can be obtained within an acceptable confidence level, modeling will proceed with the existing values.

Based on the results, answers will be sought to the following questions:

- In which countries, in which sectors, at what scale, and what types of investments can be expected in the future?
- What are the magnitudes of investment volumes?

To obtain these answers, regression, classification, clustering, time series, and panel data modeling techniques will be applied appropriately, and their reliability levels will be stated.

The results will be reported using **Power BI** and prepared as a presentation.